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RAPID COMMUNICATION

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Description of changes of key performance indicators and PRRSV shedding over time in a naïve breeding herd following a PRRS MLV exposure

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Abstract

Porcine reproductive and respiratory syndrome (PRRS) is an important economic swine disease. The usage of PRRS-modified live vaccines (MLV) is the predominant breeding herd immunologic solution used in the United States to minimize the economic losses associated with wild-type PRRS infection. Most of the current information on the effects of contemporary PRRS MLV vaccination on breeding herd performance under field conditions comes from herds with previous PRRS virus (PRRSV) exposure. Hence, there is little information on key performance indicators (KPI) changes after the exposure to a PRRS MLV in PRRSV-naïve breeding herds. The main objective of this longitudinal observational study was to describe selected KPI changes in a naïve breeding herd after PRRS MLV exposure. The secondary objective was to describe the pattern of detection of PRRSV RNA by the quantitative reverse transcriptase-polymerase chain reaction in processing fluid samples. There were transient increases for mummies during weeks 4–23 (+0.86%); increased pre-weaning mortality on weeks 3–5 (+3.76%); a decrease in live born on weeks 4–5 (–0.46) leading to a decreased pig weaned/litter on weeks 5–10 (–0.69) and increased repeated services on weeks 3–23 (+5.53%). Transient changes observed after PRRS MLV exposures did not move total pigs weaned to outside the control intervals. Starting on week 83 and for 53 consecutive weeks, there was no PRRSV detection in processing fluids, even though two whole-herd MLV exposures occurred within that period.

KEYWORDS
key performance indicators, naïve breeding herd, porcine reproductive and respiratory syndrome virus modified live vaccine, swine

1 | INTRODUCTION

Porcine reproductive and respiratory syndrome (PRRS) is a swine disease caused by the PRRS virus (PRRSV-1, PRRSV-2) (Kuhn et al., 2016; Zimmerman et al., 2019). As the name infers, PRRSV infection is associated with reproductive and respiratory disorders (Rossow, 1998; Zimmerman et al., 2019). After its description in the Netherlands and the United States, PRRSV has been detected in most pig-producing countries. Economic losses associated with PRRS were estimated to range between €3 and €160 on a per sow basis in Europe (Nieuwenhuis et al.,

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2012) and up to $663.91 million yearly in the United States swine-herd (Holtkamp et al., 2013).

Diagnosis and monitoring, biosecurity practices, herd management and immunization are measures used to control PRRS (Pileri & Mateu, 2016). Common PRRSV immunization strategies include the use of at least one exposure method: live virus inoculation (LVI), modified-live vaccine (MLV) or inactivated vaccines. The goals of immunization strategies include reducing clinical signs, viral shedding and losses in productivity associated with wild-type PRRSV infection and, consequently, reducing the economic losses associated with the disease (Pileri & Mateu, 2016). On growing animals, repeated immunization with an MLV in pigs infected with homologous wild-type PRRSV strain reduced the viral shedding at 97 days post-infection (dpi) and the number of persistently infected pigs at 127 dpi (Cano et al., 2007). Immunizing pigs with a PRRSV-2 vaccine strain and challenging them subsequently with a wild-type PRRSV-2 strain significantly reduced viral shedding in oral fluids (Linhares et al., 2012).

In the United States, there was a significant increase in the number of breeding herds exposing the breeding stock to PRRS MLV after 2012 (Perez et al., 2019; Tousignant et al., 2015). Currently, the predominant breeding herd immunologic solution used in the United States is PRRS MLV vaccines aiming to minimize the economic losses associated with wild-type PRRSV infection (Arruda et al., 2016; Murtaugh & Genzow, 2011). Breeding herds with prior exposure to a PRRSV wild-type or vaccine strain recovered the baseline productivity levels to 10 weeks before herds with no prior PRRSV exposure (Linhares et al., 2014). Additionally, the median time to consistently produce PRRSV-negative pigs at weaning (i.e. achieve stability) for herds that used MLV vaccines was 6.9 weeks longer than those using LVI as the exposure method (Linhares et al., 2014). Breeding herds classified as stable (Holtkamp et al., 2011) and exposed to a PRRS MLV showed a transient and slight increase in pre-weaning mortality (PWM) but no evidence for significant changes in other production parameters (Moura et al., 2019).

Even though exposure to PRRSV through MLV vaccination is a common intervention strategy in the United States most of the currently available information on the effects of vaccination on breeding herd performance comes from herds with previous PRRSV exposure. Hence, there is little information on key performance indicators (KPI) changes after initial exposure to a PRRS MLV in PRRSV naïve breeding herds. Availability of this information would help veterinarians and producers decide whether to immunize their herds with increasing risk of wild-type PRRSV infection. Thus, this work objective was twofold: to measure KPI changes and PRRSV detection in processing fluid (PF) over time in a naïve breeding herd after PRRS MLV exposure.

2 | MATERIALS AND METHODS

2.1 | Study design

This study was approved by the Iowa State University Institutional Animal Care and Use Committee under protocol number 19-118. This prospective field study was conducted in a commercial 6000 breed-to-wean sow herd located in the US Midwestern region from June 2018 to January 2021. Farm replacement animals were acquired from a commercial genetic source (PIC Camborough females; https://www.pic.com/) and were brought to the on-site gilt development unit (GDU) at 3–4 weeks of age. The farm facilities are composed of four barns, all connected, negative pressure, filtered ventilation and shower in-out. The farm is located in a county having a pig density of 51.39 pigs/km^2 (https://www.nass.usda.gov/Publications/AgCensus/2012/Online_Resources/County_Profiles/). The herd had no history of PRRS break and was considered naïve for PRRSV. During 10-months prior to the study, PF samples tested negative for PRRSV on ELISA and quantitative reverse transcriptase-polymerase chain reaction (RT-qPCR). Due to PRRSV activity in the surrounding geographic region, the farm owner opted to implement a PRRS MLV vaccination in an effort to develop some herd immunity for PRRSV. The study herd adopted a whole-breeding herd exposure to the commercially available Ingelvac PRRS MLV (Boehringer Ingelheim Vetmedica) by the intramuscular route using a 2-mL dose as recommended by the manufacturer.

The weekly breeding herd status for PRRSV was assessed by testing weekly PF samples for PRRSV RNA by RT-qPCR. Herd-level KPIs changes following the initial PRRS MLV exposure were accessed using a statistical process control model.

2.2 | Vaccination dates and production parameters

To stimulate the development of herd immunity for PRRSV, all breeding stock and teaser boars, independently of the production stage (lactating, gestating, weaned, replacement gilts), were exposed to the PRRS MLV on the week of 4 June 2018 (defined as week 0). Subsequent whole-herd PRRS MLV exposures occurred on weeks 17, 39, 69, 94 and 123 following a bi-annual re-exposure programme planned to re-occur in April and October. Incoming replacement gilts were exposed to two doses of a PRRS MLV in the GDU at 4 and 7 weeks after the arrival.

2.3 | Sample collection and diagnostic testing

Processing fluid samples, the serosanguineous fluid recovered from testicles and tails during processing time has been described (Lopez et al., 2018), validated (Lebret et al., 2021; López et al., 2020; Trevisan, Jablonski, et al., 2019; Vilalta et al., 2018, 2019) and increasingly used in the United States for PRRSV monitoring in breeding herds (Trevisan, Linhares, et al., 2019). Processing fluid was the sample type chosen for PRRSV monitoring in this work. PF samples were collected according to a previously described method (Lopez et al., 2018) from all processed litters by day and processing room. The PF were stored in 50 mL Falcon tubes (Fisher Scientific, Waltham, MA), frozen at −20°C, and submitted to the Iowa State University Veterinary Diagnostic Laboratory for testing. At the laboratory, the PF samples were pooled in pools of 5:1 by the week of collection and tested for PRRSV RNA by RT-qPCR and using commercial kits (Chittick et al., 2011; Pepin et al., 2015). The

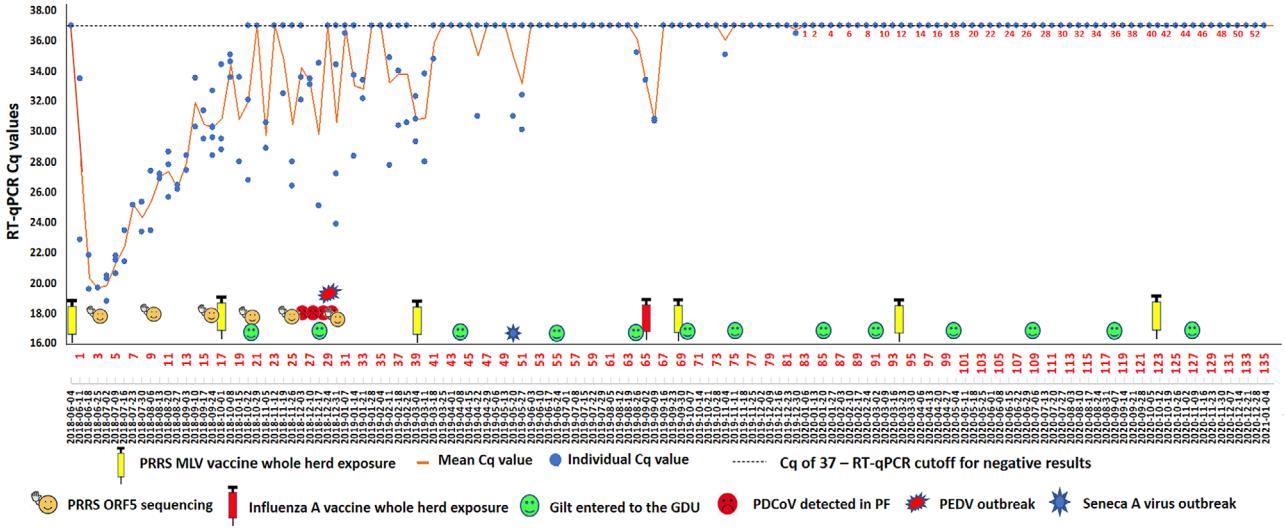

FIGURE 1 Porcine reproductive and respiratory syndrome virus (PRRSV) RNA detection in processing fluids after PRRS MLV-vaccine exposure in a naïve breeding herd. Additional events in the farm are labelled under the *X*-axis. Numbers added on the upper right Cq results indicate counts for consecutive weeks with no detection of PRRSV in processing fluids. Abbreviations: PRRS: porcine reproductive and respiratory syndrome; Cq: quantification cycle; RT-qPCR: reverse transcriptase quantitative polymerase chain reaction; MLV: modified-live virus; GDU: gilt development unit; PEDV: porcine epidemic diarrhoea virus; PDCoV: porcine delta coronavirus

results were reported as quantification cycles (Cq). When Cq was ≥37, the results were considered negative and otherwise positive. PF collected on weeks 3, 9, 16, 20, 25 and 30 and having the lowest Cq value were submitted for the open reading frame (ORF)-5 sequencing.

2.4 | Statistical analysis

Descriptive KPI analysis was implemented using an exponential weighted moving average (EWMA) available in the PROC MACONTROL procedure from SAS 9.4 (SAS Institute, Inc., Cary, NC). The EWMA parameters were set to 3 sigma's to adjust the width of upper and lower control limits, and lambda of 0.4, representing the assigned weight to the most recent variable value included in the data (Montgomery, 2009). KPI baselines were constructed based on the 42-weeks previous to the initial PRRS MLV exposure, that is two complete breeding cycles, and used to compare changes afterwards. Porcine epidemic diarrhoea (PEDV) outbreak occurred at weeks −14 and 29, porcine delta coronavirus (PDCoV) at week 26 and Senecavirus A at week 49. In function of those outbreaks, the KPIs for weeks −15 to −9 were excluded from the baseline and those from weeks 24 to 35 and from week 47 to 55 were excluded from the monitored period.

Following a previous methodology (Krieter et al., 2009; Linhares et al., 2014), a signal for significant changes in a KPI was defined as when changes were outside baseline parameters for at least two consecutive weeks. The KPIs were defined as returning to baseline when returning to the EWMA control limits and kept within parameter limits for at least a 2-week period. Monitored parameters were total pigs weaned (weaned pigs per week); total born (total pigs born per farrow); live born (number of piglets born alive per farrow); percentage of mummies (percentage of weekly mummies born from total pigs born); percentage of stillbirth (percentage of weekly stillbirth born from total pigs born); PWM (percentage of dead piglets from total pigs born); pigs weaned per litter (number of pigs per female weaned); weaning to first service interval (days from weaning to first artificial insemination service); total female deaths (number of breeding females that died or were euthanized); aborts (number of aborts per week); repeated services (percentage of weekly females returning to oestrus); and farrowing rate (percentage of females that farrowed from total serviced females within a weekly cohort).

3 | RESULTS

3.1 | PRRSV PCR detection in PF and ORF5 sequencing

The PCR Cq values were predominantly in the 'twenties for weeks 2–4 and consistently increased until week 15 (Figure 1). After the second PRRS MLV-exposure on week 17, an irregular PRRSV RNA detection pattern with negative and positive results ranging from Cq values below 30 to negative results was obtained. Following the third MLV exposure on week 39, there was no drop in Cq values below 30, and most tests were negative. Starting on week 83 and for 53 consecutive weeks, there was no detection of PRRSV RNA in PF samples, even though two whole-herd Ingelvac PRRS MLV exposures occurred within that period (week 94 and 123) (Figure 1). The six ORF-5 sequencings had genetic similarities greater or equal to 98.5% with the MLV strain used.

TABLE 1 Key parameters indicators out of control limits of detection. EWMA: Exponential weighted moving average

Key parameters indicator (process mean; standard error)	Weeks of monitoring outside the EWMA control interval	Average change during the detection of the out-of-control limit
Pre-weaning mortality (%) (13.09; 2.22)	3–5	+ 3.76
Repeated services (%) (3.75; 1.34)	3–23 63–68 117–122	+ 5.53 + 0.93 +2.96
Mummies (%) (1.08; 0.25)	4–23 73–85 90–108 112–121	+ 0.86 + 0.64 + 0.81 + 0.40
Pigs weaned per litter (heads) (11.19; 0.24)	5–10	−0.69
Live born (heads) (12.92; 0.26)	4–5 20–21 86–87	−0.46 −0.19 −0.42
Total born (heads) (14.07; 0.25)	19–20 92–95	−0.42 −0.54
Farrowing rate (%) (89.48; 2.36)	2–7 13–23 36–46 75–88 120–125	−4.38 −7.14 −8.19 −5.43 −4.67

3.2 | KPI changes

There were transient changes in KPIs with increased PWM, repeated services and mummies. Decreased KPIs were identified for live born, total born, pigs weaned per litter and farrowing rate (Table 1). No significant changes for total pigs weaned and the other monitored KPIs were detected.

4 | DISCUSSION

This prospective study investigated the detection of the out-of-control limit changes in selected KPIs of a naïve breeding herd following exposure to PRRS MLV. The observations indicated that exposing a naïve breeding herd to a PRRS MLV can potentially and transiently affect the incidence of mummies, live born, and pre-wean mortality, leading to a decrease in the pigs weaned per litter for 6 weeks (week 3–5). Most losses were concentrated during weeks 5 to 10 post the first PRRS MLV exposure. Additionally, a transient and longer increase in repeated services was observed during weeks 3–23, potentially leading to a decrease in the farrowing rate. As no detection of PRRSV occurred after week 83, the weeks with changes for different KPIs outside the monitoring control interval after week 83 could be a function of other factors not evaluated in this study and not necessarily due to PRRS activity.

The results of our study were in agreement with previous reports showing an increase in the number of mummies, decreased pigs weaned and a farrowing rate in PRRSV-negative herds after an initial MLV exposure (Baysinger et al., 1997; Dewey et al., 1999). In this study, we did not found a loss of opportunity represented by a decrease in total pigs weaned after the initial PRRS MLV exposure. Producers having breeding herds facing the potential risk of PRRSV introduction and adopting PRRS MLV practices should expect transient changes in KPIs; those will not necessarily impact total pigs weaned. A previous study has described the loss of 550 piglets as occurring from an introduction of a wild-type PRRSV strain in a herd with some previous PRRS immunity and 2457 piglets in PRRSV-negative herds (Linhares et al., 2015). As no introduction of a wild-type PRRSV strain occurred in the study herd, we could not additionally quantify the impact of such an event.

The pattern of PRRSV detection in PFs between the first and second PRRS MLV exposure agrees with previously reported results from a PRRS elimination project (Trevisan, Jablonski, et al., 2019). An intermittent PRRSV detection pattern in PF, week 17–83, was observed when some PRRSV immunity was already present in the herd. It is very likely that the introduction of new pathogens to the farm like PDCoV and PEDV at weeks 26 and 29, and Senecavirus A at week 49 may have transiently affected the herd immunity status and favouring PRRSV detection. The mass vaccination for Influenza A on weeks 64–65 was part of an annual routine exposure programme that was discontinued in 2020. No clear explanation for the drop in Ct at week 65 was encountered. It has been estimated that a preventive vaccination would be economically beneficial if the expected frequency of outbreaks were at least one in every 1 year and 9 months (Linhares et al., 2015). Even though there was no wild-type PRRSV strain introduction, the PDCoV, PEDV, and Senecavirus A outbreaks may indicate that the farm was at high risk for external agent introduction, and the MLV exposure would still be a valuable tool.

The homologous strain immunity already developed in the breeding herd has likely prevented PRRSV detection in PF after the fourth and fifth whole-herd PRRS MLV exposure on weeks 94 and 123. This agrees with reports of no detection of PRRSV-1 MLV vaccine virus in PF in stable breeding herds (Lebret et al., 2021). Additionally, when PRRSV prevalence within piglets is near zero or weak positive results are present in individual PF samples, that is high Cq values, the strategy of pooling of PF samples may fail in detecting PRRSV (López et al., 2021). Caution should be taken when using PF to monitor breeding herds that are previously exposed to homologous PRRSV MLV strain. The absence of PRRSV detection in PF may not be taken as definitive evidence of the absence of a homologous PRRSV strain in the farm.

This study provided valuable information to assist producers and veterinarians in designing intervention strategies for PRRS naïve herds facing PRRSV introduction risk. A limitation of this study was that it followed only one PRRS naïve breeding herd exposed to one commercially available PRRS MLV and may not represent variation in PRRSV detection and KPI changes across different animal populations. Moreover, the whole-herd exposure was to a single PRRSV-2 MLV and may not reflect the response when a different PRRS MLV is in use. Additionally, no wild-type PRRSV strains were detected on the farm that could affect the PRRSV pattern of detection.

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CONFLICT OF INTEREST

The authors declare no conflicts of interest.

ETHICAL STATEMENT

The authors confirm that the ethical policies of the journal, as noted on the journal's author guidelines page, have been adhered to. This study was approved by the Iowa State University Institutional Animal Care and Use Committee under protocol number 19-118.

DATA AVAILABILITY STATEMENT

Available data was provided in the manuscript; additional farm data is unavailable due to producer confidentiality.

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